
Computational Fluid Dynamics Uses in Fluid Dynamics/ Aerodynamics Education

Terry L. Holst, Ames Research Center, Moffett Field, California

July 1994



Ames Research Center
Moffett Field, California 94035-1000

Computational Fluid Dynamics Uses In Fluid Dynamics/Aerodynamics Education

TERRY L. HOLST

Ames Research Center

Summary

The field of computational fluid dynamics (CFD) has advanced to the point where it can now be used for the purpose of fluid dynamics physics education. Because of the tremendous wealth of information available from numerical simulation, certain fundamental concepts can be efficiently communicated using an interactive graphical interrogation of the appropriate numerical simulation data base. In other situations, a large amount of aerodynamic information can be communicated to the student by interactive use of simple CFD tools on a workstation or even in a personal computer environment. The emphasis in this presentation is to discuss ideas for how this process might be implemented. Specific examples, taken from previous publications, will be used to highlight the presentation.

Introduction

Use of computational fluid dynamics (CFD) tools by the aerospace field to increase understanding of fluid dynamic and aerodynamic phenomena has been rapidly increasing during the past decade especially the last several years. The primary reasons for this are the rapidly increasing simulation capabilities in the CFD field and the rapidly expanding capabilities in computer hardware performance. For example, computer hardware execution speed has increased by a factor of about 15 over the past decade and by over 200 during the past two decades. This rapid advance in computational execution speed is displayed in figure 1. The top curve shows how the theoretical peak execution speed has improved with time and includes effects from both circuit speed and architectural improvements, e.g., vectorization speed ups from Cray-type computers. The lower curve represents the improvement in execution speed due to just circuit speed. The middle curve represents the actual improvement in execution speed performance from a variety of CFD application codes as approximated from the shaded symbols.

Another reason for the dramatic increase in the use of scientific computational tools is that industry has discovered the positive influence that computational analysis can

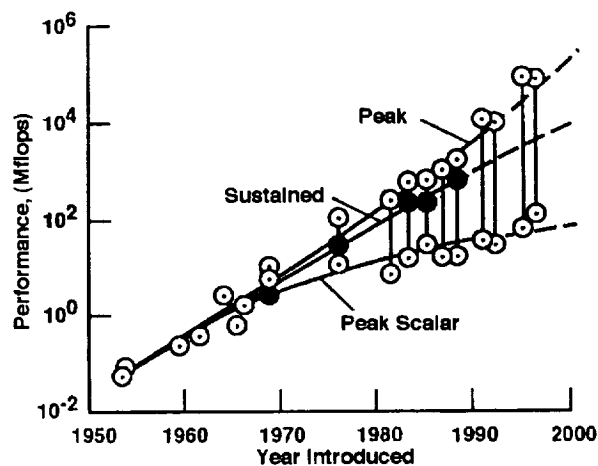


Figure 1. Conventional main-frame computer execution speed improvement as a function of time, results taken from references 1-4.

have on aircraft, spacecraft, and missile design. Improved efficiency in aerospace vehicle performance at reduced design cost and risk is a direct result of increased use of computational simulations. Indeed, additional advancement in this area is crucial to enable the United States to maintain its technological advantage in the aerospace sciences.

Just as numerical simulation has become a significant and growing aspect of the aircraft design process, the stage is set for a dramatic increase in the utilization of CFD in the educational arena. In this context, it is not meant to imply that the study of CFD will increase dramatically, but that the use of CFD as a teaching tool for other areas or disciplines of fluid dynamics will increase. This utilization should range from enhancing the understanding of nonlinear engineering models, e.g., the aerodynamics of transonic wings, to obtaining a better understanding of fluid physics, e.g., flat plate boundary layer transition. Through the synergistic utilization of CFD coupled with an appropriate level of experimental validation, students will obtain a better understanding of the physical aspects of aerodynamics and fluid mechanics as well as how to

interpret the effects of numerical error associated with CFD solutions.

The remainder of this paper will provide a review of some of the current areas of CFD research that may be utilized in the educational environment in the near future. These areas are especially attractive if improvements in workstation computing continues at today's current fast rate. In addition, areas that may be used more immediately, i.e., even today, will be presented and discussed.

Review of CFD Applications

The first results used to establish the abilities of CFD are a set of full potential solutions for a variety of transonic wing configurations (taken from refs. 5-7). In these simulations the nonlinear full potential equation is solved for the inviscid transonic flow field complete with transonic shock waves. Results from two different full potential computer codes, TWING (ref. 5) and FLO28 (ref. 6), are compared with experiment (ref. 7) in figure 2 for the ONERA M6 wing at transonic flow conditions. Although there are some discrepancies in the computed results, both show the same trends, i.e., they both predict a double shock structure on the upper surface including a supersonic-to-supersonic oblique shock swept

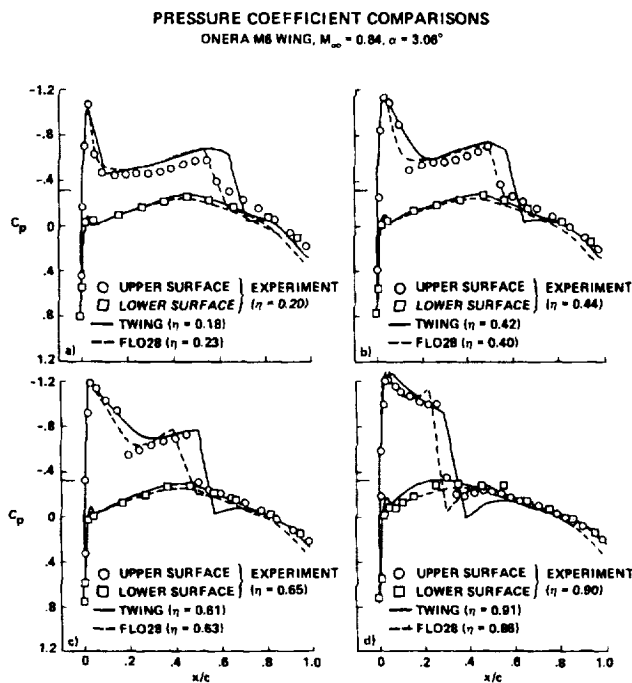


Figure 2. Pressure coefficient comparisons at four semi-span stations on an ONERA M6 wing, $M_\infty = 0.84$, $\alpha = 3.06^\circ$ (ref. 5).

approximately parallel to the wing leading edge. Most of the discrepancies are a direct result of a coarse grid used in the numerical simulations, a direct result of main memory limitations from a decade ago.

An additional full potential result computed with the TWING code and taken from reference 8 is shown in figure 3. In this figure the drag-rise characteristics (C_D versus M_∞) are compared for two transonic wing cases: an original or baseline wing and an optimized wing. The baseline geometry was modified using the QNMDIF optimization code (refs. 9-10) to produce the optimized wing by minimizing the value of cruise D/L. As can be seen from figure 3 the drag-rise characteristics of the optimized wing are significantly improved over the original baseline wing. It should be pointed out that the drag values associated with figure 3 are pressure drag values only, i.e., they do not contain skin friction drag.

The most interesting aspect of these simulations, especially in the present context, is the amount of computer time required for a complete simulation. The computing times reported in reference 5 are on the order of 10-20 sec on a single processor of a Cray Y-MP computer. An entire aerodynamic performance curve, such as the drag rise curve presented herein, requires on the order of only one minute. This machine was in the supercomputer class just a decade ago but now is about even with a high end workstation in execution speed and does not even match a reasonably advanced personal computer in terms of main memory. Such simulations would be easily adapted to the educational environment and will be discussed in more detail in the next section of this paper.

The next example results used to establish the state of the art in CFD applications are a set of Reynolds-averaged

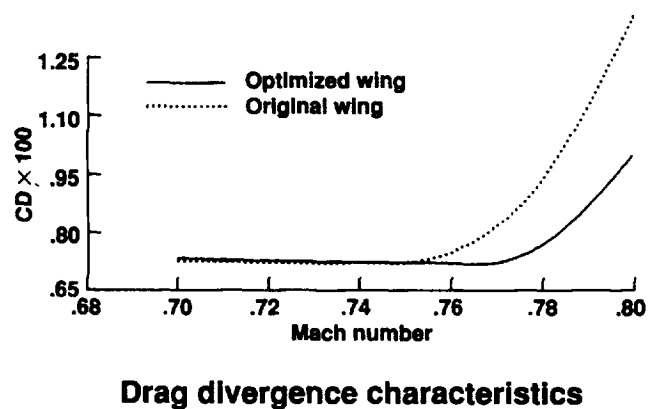


Figure 3. Coefficient of drag ($\times 100$) versus Mach number for baseline and optimized wings (ref. 8).